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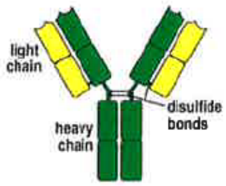
1. Introduction

Immunoglobin (Ig) is a globulin with antibody activity or chemical structure similar to antibodies. It has tetrapeptide structure consist of two same light chains and two same heavy chains, they from tetrapeptide structure via disulfide bonds.



Heavy chain are divided into five categories according to antigenic difference, if heavy chain is γ, the Ig will be IgG. α chain corresponds to IgA, μ chain corresponds to IgM, σ chain corresponds to IgD and ε chain corresponds to IgE.

Research shows that intravenous immunoglobulins (IVIG) can be used for immunomodulatory. IVIG is composed of more than 95% of IgG, no more than 2.5% of IgA, little IgM, soluble CD4 and CD8, human leukocyte antigen molecules and certain cytokines. IgG subtype composition is basically consistent with normal human serum, which IgG is about 55% to 70%, IgG2 is about 0% to 6%, IgG4 is about 0.7% to 2.6%. Because it is rich in broad-spectrum antiviral, antibacterial and anti-pathogen IgG antibodies, the immunoglobulin idiotypic and idiotype antibodies will form a complex immune network, with dual therapeutic effects of immune substitution and immune regulation.

2. Current Progress

For immunodeficiency patients, the role of IVIG is to supplement the deficiency of immunoglobulins. However, the mechanism of the powerful immunomodulatory and anti-inflammatory effects of IVIG in the treatment of autoimmune diseases has not yet been fully clarified. Most IgGs in IVIG exist as monomers, and are composed of two parts: Fab segment and Fc segment: Fab segment binds to specific antigen and participates in specific immune response; Fc segment participates in non-specific immune response. Both of the above functional regions are involved in the immune regulation of IVIG, involving various levels of the immune system and parts of the inflammatory response. Specifically include 8 parts. First, Fc receptor-dependent immunomodulatory effects, including blocking activated FcγR, can significantly inhibit phagocytosis of sensitized cells in the mononuclear macrophage system (MPS) and antibody-dependent cytotoxicity (ADCC). It effects Activating and / or inhibiting the expression of FcγR, inhibiting the function of macrophages, exerting anti-infective effects. Also it increases the catabolism of endogenous IgG, and promoting the elimination of pathogenic autoantibodies in the body. Second, Immunomodulatory effects of anti-idiotypic antibodies. There are many anti-idiotypic antibodies in IVIG, which can bind to the variable region of autoantibodies in the body and act as a neutralizing antibody. At the same time, they can also bind to B-cell surface receptors and inhibit the activation, proliferation and Production of autoantibodies. This immunomodulatory effect of IVIG has a slow and long-lasting anti-infective effect. Third, inhibit complement-mediated inflammatory response. This immunomodulatory effect of IVIG through the Fc segment is rapid and long-lasting. Fourth, Regulating effect on dendritic cells (DC). IVIG low-dose replacement therapy can promote the differentiation of DCs, while high-dose administration can inhibit the maturation and differentiation of DCs and promote the apoptosis of DCs, thereby inhibiting the activation and proliferation of autoreactive T cells. Fifth, inhibition of B cells and antibodies. High-dose IVIG can directly neutralize autoantibodies or inhibit B cell proliferation through anti-idiotypic antibodies, reduce the production of autoantibodies, and thus treat various autoimmune diseases mediated by autoantibodies. Sixth, inducing apoptosis. IVIG can induce apoptosis of lymphocytes and monocytes through the Fas-FasL pathway, and can also treat Kawasaki disease (KD) by inducing neutrophil apoptosis. Seventh, impact on cytokine networks. IVIG can regulate the activity of T, B, DC and monocyte macrophages, and directly or indirectly affect the secretion and release of cytokines. Eighth, effect on chemotaxis of cells. IVIG can affect the expression and role of chemokines or adhesion molecules, affect the recruitment of inflammatory cells, and regulate the inflammatory response.

Based on good clinical controlled trials, the United States FDA has approved six types of diseases for IVIG treatment. They are ITP, primary immunodeficiency, secondary immunodeficiency, HIV infection in children, KD, prevention of graft-versus-host disease (GVHD) and infection after bone marrow transplantation (BMT). The indications for IVIG use approved in Italy are three parts. First is immunodeficiency patients, including children with primary, secondary, and AIDS. Second is immunomodulation, including ITP, Guillain-Barre syndrome (GBS), and KD. Last is treatment and prevention of allogeneic BMT infection and GVHD, insufficient antibody production.

The vast majority of IVIGs are used in immunomodulatory therapies. Darabi et al. summarized all the 194 cases of IVIG infusion approved by the Ad Hoc Committee of the Massachusetts General Hospital in 2004. These patients shared 230 g of IVIG 48 , and were ranked according to the number of patients with chronic neuropathy, including chronic inflammatory demyelinating polyneuropathy (CIDP), multifocal motor neuropathy (MMN), and secondary hypogammaglobulinemia, ITP, primary hypogammaglobulinemia, renal transplant rejection, myasthenia gravis (MG), GBS, necrotizing fasciitis, autoimmune hemolytic anemia (AIHA), KD, dermatomyositis , H1V, liver transplantation anti-rejection, desensitization before heart transplantation, myopathy, juvenile rheumatoid arthritis, parvovirus infection, neonatal hemolytic disease (HDN), and neonatal immune thrombocytopenia.

Liumbruno et al. concluded that IVIG is routinely used in clinical treatment of the following diseases. Hematological diseases, including neonatal immune immune thrombocytopenia, AIHA, HDN, immune-mediated neutropenia, purpura after blood transfusion, pure Red blood cell aplastic anemia (PRCA) and platelet transfusion are ineffective, Infectious diseases, namely prevention of cytomegalovirus (CMV) infection after solid organ transplantation, Nervous system diseases, including acute diffuse optic neuromyelitis, CIDP , Refractory children with epilepsy, myasthenic syndrome, Lambert-Eaton syndrome (LEMS), MMN, multiple sclerosis

(MS) and stiff-person syndrome,rheumatism, including skin muscles Inflammation, polymyositis, systemic lupus erythematosus (SLE) and systemic vasculitis and Desensitization before kidney transplantation. Although these usages have been reported to be effective and some have formed expert consensus, due to the lack of large series of studies and low level of evidence, large-scale clinical randomized controlled trials (RCTs) are still needed for verification.

In addition, there are a large number of clinical cases of unreasonable use of IVIG like Hematological disease, including acquired hemophilia A, acquired vascular hemophilia, aplastic anemia (AA), and congenital pure red Aplastic anemia (DBA), thrombotic thrombocytopenic purpura (TTP), and hemolytic uremic syndrome Second is infectious disease, including burns (prevention of infections), HIV infection (adults), surgery and trauma (prophylactic). Third is rheumatism, including inclusion body myositis (IBM) and rheumatoid arthritis (adolescent and adult). Last is other disease, such as acute cardiomyopathy, acute idiopathic familial autonomic dysfunction, acute Lymphocytic leukemia, acute renal failure, white matter adrenal atrophy, amyotrophic lateral sclerosis, autism, diabetes, diabetic neuropathy, and POEMS syndrome. From current clinical evidence, IVIG is ineffective in treating these diseases.

3. Challenges

challenges include side effect what else?

As a ~~currently~~ widely used plasma in the world, IVIG-scale human plasma has a complicated preparation process, tight supply, and expensive, but also has some adverse reactions and potential adverse consequences. IVIG is relatively safe, with the incidence of adverse reactions ranging from 1% to 15%. Most adverse reactions are mild, and severe adverse reactions are rare but fatal.

IgG polymer and complement activation

The syndrome includes headache, chills, high fever, allergies, nausea, vomiting, joint pain, and hypotension associated with anaphylactic shock. The reasons may be related to the activation of IgG polymers, IgG dimers, and the complement pathway. These polymers can even activate complement in the absence of antigen. Discontinuation of IVIG or slowing the drip rate can resolve symptoms. Use of aspirin, acetaminophen, antihistamines, or glucocorticoids can prevent the syndrome.

Hypotension

This adverse reaction is rare, manifested as a sudden drop in blood pressure during IVIG infusion, and may be related to the presence of IgG dimers in the product. Unlike IgG polymers, IgG dimers do not activate complement, but can act on blood pressure. Another reason may be related to the activation of macrophages, monocytes and neutrophils, followed by the release of platelet activating factors.

Impairment of renal function

The exact mechanism is not clear, but from the evidence of renal histopathology, it is related to the sucrose stabilizer present in IVIG products. Sucrose cannot be directly absorbed. It must be hydrolyzed to glucose and fructose in the small intestine before it can enter the portal circulation to be transported to the whole body. It cannot be hydrolyzed if it is administered intravenously. When excreted through the kidney, it can cause renal tubular permeability damage. Patients with advanced age, diabetes, hypovolemia, sepsis, hyperglobulinemia, renal insufficiency, and concurrent use of nephrotoxic drugs should be used with caution. If they must be used, IVIG products with low sucrose content should be selected. or dilute with distilled water and slow down the drip rate.

Aseptic meningitis

It usually occurs 6 to 48 hours after IVIG infusion, with severe headaches, fever, nausea and vomiting, neck stiffness, drowsiness, and photophobia, while the levels of lymphocytes and proteins in the cerebrospinal fluid increase. The adverse reaction is dose-related, and it does not occur at low doses, and the incidence rate can reach 10% at high doses, especially in patients with a history of migraine.

Thromboembolic

Including deep vein thrombosis, myocardial infarction, pulmonary embolism, central retinal vein occlusion, stroke, and fatal hepatic vein occlusive disease adjacent to the injection site, the mechanisms include IVIG increased blood viscosity, increased platelet number and platelet activation accumulation, and presence in IVIG The trace coagulation factors, especially FXIa, cause thrombosis, vasospasm, and increased red blood cell aggregation. Old age, overweight, previous cerebral infarction or myocardial infarction, and prolonged bed rest are high-risk factors.

Allergic reactions in people with IgA deficiency

IVIG products contain a small amount of IgA. IgA-deficient people will produce anti-IgA IgG and IgE antibodies after importing them. After re-entering IVIG, serious or even lethal allergic reactions may occur, especially in patient who has anti-IgA IgE antibodies. Therefore, patients with a history of severe allergies should choose IVIG products with low IgA levels.

Other adverse reactions

Includes volume overload and pulmonary edema, transfusion-related acute lung injury (TRALI), skin eczema, arthritis, hemolytic anemia, leukopenia, and virus-transmitted diseases, etc. when used in large doses.

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